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Exploring DEG calling in the presence of limited patient samples

Introduction

For detection of differentially expressed genes (DEGs) in RNA-sequencing data, approaches such as *DESeq2* **[1]** apply statistical tests to compare two conditions. Although accurate statistics always call for large group sizes, resources are often limited and researchers can only offer a limited number of samples.

Exploring current constraints of DEG calling in the context of limited data, we evaluated the performance of DESeq2 at a low number of samples being analyzed. DESeq2 is a comparatively sensitive tool that can find DEGs even in the presence of only 3 samples **[3]**. To increase the ratio of true positives we propose a strategy to re-evaluate DEG calls by intersecting the discoveries with results of a second analysis that is performed on a subset of the original samples.

Methods

To base our analysis on a set of validated DEGs, we synthetically generated count data using the *compcodeR* R package **[2]**. We simulated data for low numbers of samples, ranging from 2 to 10 samples per group. For each sample size a total of 88 sets of count data with varying parameters were generated:

- Total number of genes: 10000
- Differentially expressed genes: $5\%|10\%|30\%|50\%$
- Outlier chance per gene: 5%
- Effect sizes: $X \sim Exp(1) + \lambda,\ \lambda \in [0.5, 1.5]$

Differential expression analysis was performed using the *DESeq2* R package with standard settings and no pre-processing. We applied a FDR-control of 5%.

Results

Performance on limited data

To estimate performance we calculated precison and recall by comparing the DEGs called by *DESeq2* with the ground truth of the simulated data. Figure 1 compares the performance of group sizes ranging from 1 to 10. We observe that recall steadily increases with the number of samples, but high precision can be retained, depending on the fraction of true DEGs. For instance, data featuring high proportions of true DEGs ($> 30\%$), already reach maximum precision at four to five samples per group.

Recall
steadily increases along with the number of samples.

Precision
reaches its maximum at a relatively small group size, depending on the fraction of true DEGs.

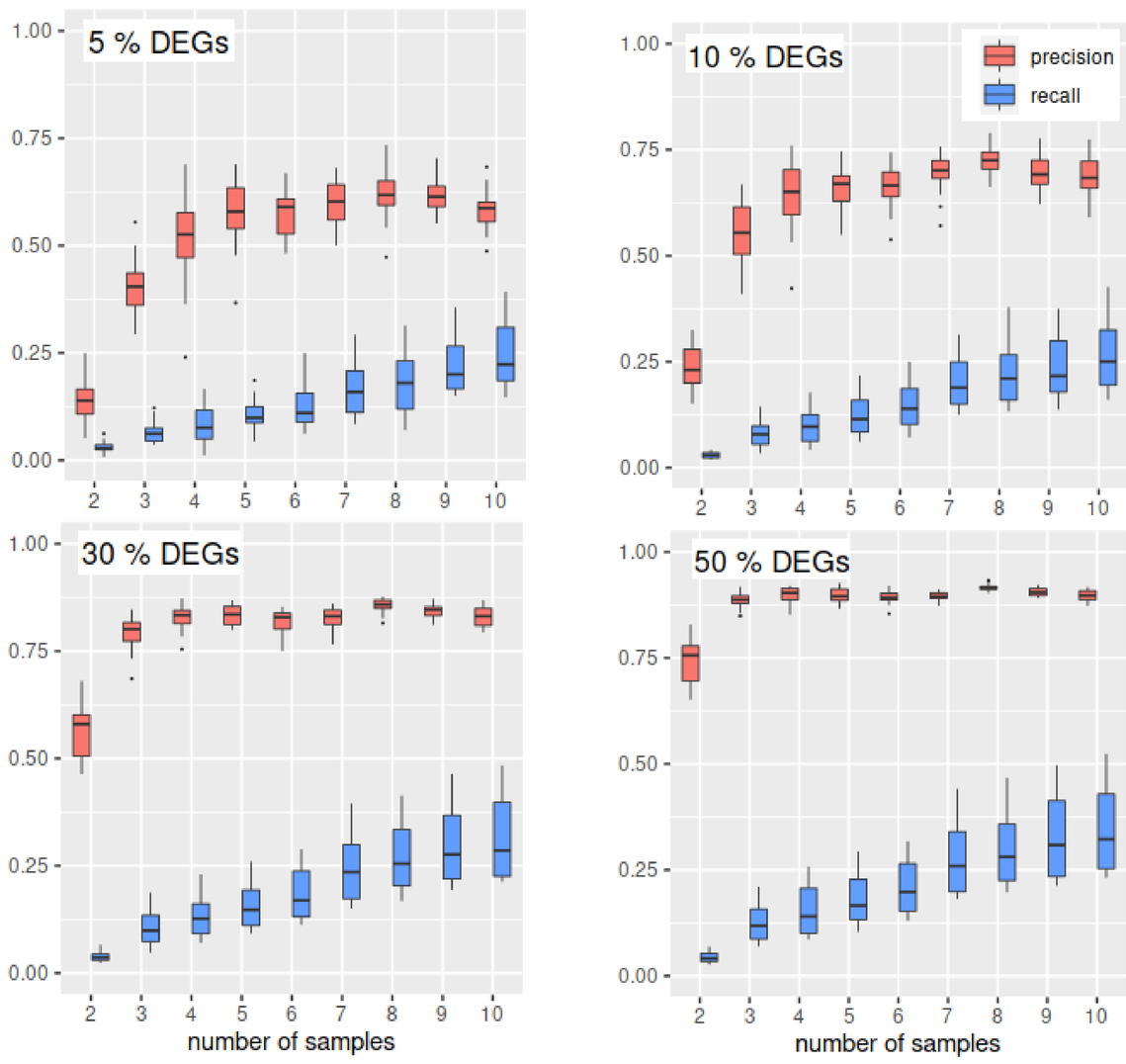

Figure 1: Comparison of performance of DEG calling results based on varying fraction of true DEGs (% DEGs) and varying number of samples per group.

Unequal group sizes

Sometimes samples for one of the conditions are especially hard to aquire. Researchers might want to compensate by providing a large reference group. This strategy however only moderately increases recall, as shown in Figure 2. It would be more favourable to slightly increase both group sizes, even if the total number of samples were lower.

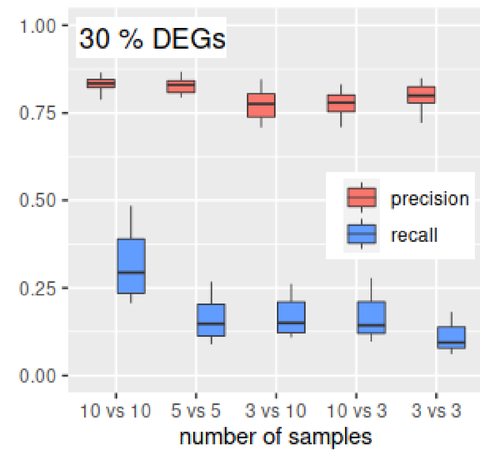

Figure 2: Performance of DEG calling results including unequal group sizes.

Re-evaluating DEG calls using data subsets

Because of low recall, researchers should not hope to gain extensive insight over all differentially expressed genes and networks. Instead, one can search for candidate genes to investigate in later studies. To avoid selecting false positives, DEG calls can be evaluated by their p-value, effect size or gene function. We further propose a stratey to re-evaluate genes by using results from data subsets:

Additionally to calling DEGs on the original group size of n samples per condition, we performed a second analysis on a subset of $n-1$ samples each. Then, only genes that were called in both analyses (intersection) were evaluated. Figure 4 shows that this procedure increases precision and can thus help to identify true positive candidate genes.

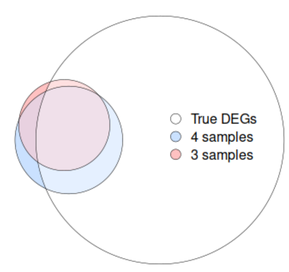

Figure 3: Exemplary visualization of intersecting DEG calling results of $n = 4$ and $n - 1 = 3$ samples per group.

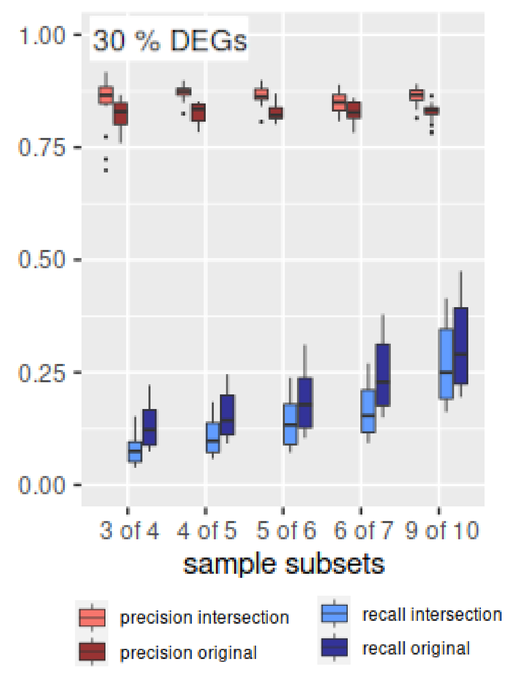

Figure 4: Comparison of performance of 'original' and 'intersection' DEG calling results. '3 of 4' refers to an original group size of $n = 4$ per group.

Conclusion - Working with limited data

- Consider level of difference between conditions (estimated % DEGs).
- Take additional steps to improve recall, such as filtering low-expression genes.
- Focus resources on increasing size of the smaller group (if possible).
- Aim for explorative research to generate hypotheses.

References

[1] Love MI et al. Moderated estimation of fold change and dispersion for RNA-seq data with DESeq2. *Genome Biology* 2014, 15,550.

[2] Soneson C. compcodeR - an R package for benchmarking differential expression methods for RNA-seq data. *Bioinformatics* 2014, 30(17),2517-2518.

[3] Baik B., Yoon S., Nam D. Benchmarking RNA-seq differential expression analysis methods using spike-in and simulation data. *PLoS ONE* 2020, 15(4):e0232271.

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